

Chapter 7

Fixed Effects, Random Effects, and Hybrid Models for Causal Analysis

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Abstract Longitudinal data are becoming increasingly common in social science research. In this chapter, we discuss methods for exploiting the features of longitudinal data to study causal effects. The methods we discuss are broadly termed *fixed effects* and *random effects* models. We begin by discussing some of the advantages of fixed effects models over traditional regression approaches and then present a basic notation for the fixed effects model. This notation serves also as a baseline for introducing the random effects model, a common alternative to the fixed effects approach. After comparing fixed effects and random effects models – paying particular attention to their underlying assumptions – we describe hybrid models that combine attractive features of each. To provide a deeper understanding of these models, and to help researchers determine the most appropriate approach to use when analyzing longitudinal data, we provide three empirical examples. We also briefly discuss several extensions of fixed/random effects models. We conclude by suggesting additional literature that readers may find helpful.

Introduction

The problem of causal inference is fundamentally one of unobservables. – Charles Halaby (2004: 508)

Fixed effects models provide a way to estimate causal effects in analyses where units (individuals, schools, neighborhoods, etc.) are measured repeatedly over time. The beauty of the fixed effects method is that it can eliminate the effects of confounding variables without measuring them or even knowing exactly what they are, as long as they are stable over time. In this chapter we describe the kinds of confounding variables that the fixed effects method eliminates and demonstrate how the method does so. We also compare the fixed effects method to its closest cousin and chief alternative, the random effects method, and discuss recent hybrid methods that combine appealing features of the two methods.

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The chapter unfolds as follows. The first section establishes essential notation for the fixed effects model. From this notation, it is relatively easy to demonstrate the advantage of fixed effects models over more conventional regression methods for dealing with omitted-variable bias. The second section introduces the random effects model and compares it with the fixed effects model, highlighting the key differences across the models. The third section describes a hybrid model that contains features of both the fixed effects and random effects models. We follow this by summarizing the guiding principles for choosing between fixed effects and random effects models. We then provide empirical illustrations of the different models, including tests that allow researchers to compare fixed and random effects coefficients and models directly. We conclude by briefly outlining extensions of the fixed effects model to different types of data and response variables and provide annotated references for those who are interested in further reading on fixed effects, random effects, and hybrid models for analyzing longitudinal data.

The Fixed Effects Model

To understand the fixed effects model, it is useful to begin with the linear regression model for quantitative dependent variables. The standard regression model can be expressed in the following generic form, where $\beta X = \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_K X_K$:

$$Y_i = \alpha + \beta X_i + \varepsilon_i \quad (7.1)$$

Y_i is the value of the dependent variable for the i th unit, α is an intercept, β is a row vector of regression coefficients, X is a column vector of the K causes of Y , and ε is a random disturbance. We assume that ε has a mean of zero and constant variance and that it is uncorrelated with the variables in X . We also assume (for now) that Y is a quantitative variable.

Estimation of the standard regression model is complicated by the possibility of omitted-variable bias resulting from the failure of the researcher to include all possible K causes of Y in the empirical model. The size of the bias depends on the magnitude of the effects of the omitted causes and the strength of the correlation between the measured and unmeasured causes. The penalty paid for the omission of causal variables can be severe (Leamer 1983). Indeed, because of its pervasiveness and the danger it poses to the estimation of causal effects, omitted-variable bias is a transcendent issue in social science methodology, a problem that is particularly acute with regard to nonexperimental research designs (our focus here). Method texts would be much shorter, and estimation procedures much simpler, if it were always easy to identify all causes of Y and to measure them without error.

With these points in mind, we restate Eq. (7.1), this time distinguishing the measured (X) and unmeasured (X^*) causes of Y :

$$Y_i = \alpha + \beta X_i + \beta^* X_i^* + \varepsilon_i \quad (7.2)$$

In this equation, unlike (7.1), the vector X includes only a subset of the causal variables; the unmeasured causes are in vector X^* . Because some of the causal variables are missing, the regression of Y on X will generally result in biased estimates of the coefficients in β .

Consider how we might alleviate this bias. The most common approach in nonexperimental research is to try to reduce the size of the vector X^* , that is, to try to measure all the causes we can. Assuming we can measure at least some of those causes, we have several options: We can add them directly as predictors in the standard regression model, we can use them as tools to match individual units (as in propensity score methods), or we can use both matching and regression (see Ho et al. 2007).

In social science research, however, it is virtually impossible to identify and measure all causal variables. It is sometimes possible nevertheless to alleviate omitted-variable bias by reducing or eliminating the association between the unmeasured causes and the cause or causes that we are interested in. One of the most common methods is through random assignment. Although random assignment figures most prominently in experiments in the social sciences, randomization provides the basis for other social science methods as well, such as the use of instrumental variables (Firebaugh 2008, chapter 5). The attractive feature of randomization is that it can alleviate the biasing effects of confounding variables without actually measuring them (or even knowing what they are).

Random assignment is not, however, the only tool social scientists have for alleviating the confounding effects of unmeasured causes. Indeed, since many social scientists use secondary data, randomization might not even be an option. However, if the data under consideration are longitudinal, the fixed effects approach can also alleviate the effects of confounding variables without measuring them (Allison 2005; Halaby 2004; Hsiao 2003; Wooldridge 2010) – and it can do this even in the absence of random assignment. To show how this is possible, we first revise Eq. (7.2) to represent an analysis based on panel data with $t = 1, 2, \dots, T$ measurements for each unit:

$$Y_{it} = \alpha_t + \beta X_{it} + \beta^* X_i^* + \beta^{**} X_{it}^{**} + \varepsilon_{it} \quad (7.3)$$

By modeling different intercepts for each point in time, the term α_t allows for period effects (those not accounted for by other aspects of the model) that change the response variable by the same amount for each unit. And by appending the subscript “ t ” to the appropriate “ i ” in Eq. (7.2), we are permitting values to change for the i th unit over time. Thus, for example, Y_{it} denotes the value of Y for unit i at time t .¹

The critical point of this new notation is to subdivide the unmeasured causes into those that change over time (the X_{it} variables) and those that are constant over time (the X_i variables). For the latter, we include the subscript i , to indicate that the causes vary from unit to unit, but not the subscript t . (For convenience, we often assume below that the units are persons, but they could be neighborhoods, ethnic groups, classrooms, nations, corporations, and so on.) Date of birth, for example, varies across individuals, but is a constant for any given individual. Educational attainment, by contrast, varies both across persons and over time because individuals can add to their education. The vector of time-invariant causes (X_i) often includes important causes that are hard to measure and hence are likely to remain unmeasured in many analyses, such as individual-level personality traits, or geographic features in the case of nations.

The β s in Eq. (7.3) have no subscript t , meaning that the coefficients are constant over time. Hence, for each of the stable unmeasured causes X_i , the product $\beta^* X_i^*$ is constant over time. It follows that the *sum* of these products is also constant over time for persons, although it varies across persons. In other words, $\beta^* X_i^* = \beta_1^* X_{i1}^* + \beta_2^* X_{i2}^* + \dots + \beta_i^* X_{ip}^* = \mu_i$, where μ_i is a constant for the i th individual.

Substituting μ_i for $\beta^* X_i^*$ in Eq. (7.3), we have:

$$Y_{it} = \alpha_t + \mu_i + \beta X_{it} + \beta^{**} X_{it}^{**} + \varepsilon_{it} \quad (7.4)$$

If all the unmeasured causes are constant over time, then X_{it}^{**} is empty by definition, and Eq. (7.4) reduces to:

$$Y_{it} = \alpha_t + \mu_i + \beta X_{it} + \varepsilon_{it} \quad (7.5)$$

¹ Because time is nested in the units, it might seem more natural to use the notation Y_{it} instead, as in Raudenbush (2009). We nonetheless follow convention in the fixed effects literature and place the subscript “ i ” before the “ t .”

Equation (7.5) is the foundational fixed effects model. It is sometimes referred to as a “two-way fixed-effects model” because it allows for both period-specific (α_t) and unit-specific (μ_i) fixed effects. The key is the term μ_i , a term that varies across persons but is constant for each person over time (note that the term has no subscript for time). This term, then, captures all relevant differences between persons that are stable over time and are not accounted for by the other independent variables in the model. Because we have multiple observations for each person, we have sufficient degrees of freedom to include dummy variables for the μ_i . Hence, one common way to estimate a fixed effects model is with person-specific dummies. Observe that the dummy variables remove the stable effects (constant β s) of time-invariant unmeasured causes (constant X_i s) even when we do not know what those causes are. As a result, the fixed effects approach is less prone to bias because its assumptions about unmeasured causes are more realistic than the assumptions that we usually need to make about those causes. We refer to this as the *fixed effects advantage* (Firebaugh 2008, p. 136):

Fixed effects advantage. The fixed effects approach replaces the typically unrealistic assumption that the measured and unmeasured causes are uncorrelated with the less restrictive assumption that the unmeasured causes are constant, and their effects are stable.

In sum: The fixed effects approach removes the effects of time-invariant causes, whether those causes are measured or not. That is a powerful feature because it means that fixed effects methods can alleviate omitted-variable bias in a less-than-fully-specified model. Moreover, fixed effects models eliminate this bias even when we do not know what a fully specified model would look like. It is important to keep in mind, however, that the fixed effects approach does *not* remove the biasing effects of time-varying confounders, so the key assumption of the method is that unmeasured causes are constant.

Fixed effects models are especially useful, then, in instances where there are important causes of Y that are hard to measure and that tend to change slowly, or not at all, over time, a common situation in nonexperimental research. For research on individuals, difficult-to-measure enduring causes might include personal traits such as ambition, work ethic, determination, or physical attractiveness. In research on crime, it might be important to control for individual differences in the propensity to commit crime, yet that is hard to measure directly. Likewise, in studies of corporations, “corporate culture” is an important, yet difficult-to-measure, concept. In the case of cross-national research, there are many examples of country characteristics that are slow to change yet difficult to measure with standard social science data – factors such as a country’s history, geographic location, and access to seaports. As these examples illustrate, across the social sciences, the most troublesome unmeasured causes very often are slow-to-change, hard-to-measure traits. Thus, the ability of fixed effects models to alleviate the biasing effects of such causes is a very attractive feature.

What Is Sacrificed in Fixed Effects Models?

As Allison (2005: 3) observes, “The essence of a fixed effects method is captured by saying that each individual serves as his or her own control . . . [which is] accomplished by making comparisons *within* individuals.” As is often the case with estimation methods, model strengths and weakness are often related. While personal comparisons are powerful, a great deal of information can be lost by focusing only on variation within individuals, thereby ignoring the variation across individuals. Because fixed effects models remove the effects of all time-invariant causes – measured as well as unmeasured – the standard fixed effects model is unable to estimate the effects of time-invariant *measured* causes. Whether this presents a problem for researchers is, of course, related to the substantive questions of interest. If the causes of interest are time-invariant, standard fixed effects is not appropriate (this

limitation often can be overcome, however, by using hybrid models [described later] or instrumental variables). In contrast, if the causes of interest are all time-varying, then the inability to estimate time-invariant predictors becomes inconsequential.

The fixed effects focus on within-unit variance – the feature that makes it attractive – also reduces its statistical power. As estimated in the fixed effects method, the μ_i -term is an ignorance term, that is, it is a fitted value for each unit that reflects unit differences without indicating why the units are different. Although this strategy makes fixed effects an effective method for alleviating the confounding effects of unmeasured time-invariant causes, it involves the loss of information. Hence, the fixed effects method is often less efficient than estimation methods that are based on between-unit variance as well as on within-unit variance.

The primary alternative approach to estimating causal effects with panel data is the random effects method which, unlike the fixed effects method, makes use of between-unit as well as within-unit variance. While the random effects method potentially can deliver more powerful tests of hypotheses and narrower confidence intervals, it is more vulnerable to omitted-variable bias from unmeasured time-invariant causes. We now consider the random effects model in more detail.

The Random Effects Model

For the random effects model, we start with Eq. (7.5), the same as the foundational equation for the fixed effects model:

$$Y_{it} = \alpha_i + \mu_i + \beta X_{it} + \varepsilon_{it} \quad (7.6)$$

The difference between random effects and fixed effects boils down to this question (Nerlove 1971): Do we treat μ as a random variable? In both models, μ_i represents the composite effect of unmeasured traits that vary across individuals. In the fixed effects approach, we treat the μ_i as fixed constants and omit all between-person variance from the model. The random effects approach, on the other hand, declares the μ_i to be drawn from a random variable μ with fixed variance that can be estimated from the data.

The random effects and fixed effects approaches differ, then, in the way they estimate the individual-specific term (Hsiao 2003; Wooldridge 2010). The random effects approach treats this individual-specific effect as randomly varying, whereas the fixed effects approach treats it as fixed for each individual. The usual assumption of the random effects model is that μ has a zero mean and constant variance and that it is independent of the X s and of ε_{it} . We also assume, as we did for the fixed effects model, that the β s are constant over time (though they might vary across units: see below) and that ε_{it} is a zero-mean error term that is independent of the X s as well as with the random variable μ .

The random effects method can be thought of as regression with a random constant term for the person-specific intercepts (Greene 2003). That is, one way to handle the ignorance term represented by μ is to assume that the intercept for each person is a random variable consisting of a mean value plus a random error. For this model to provide unbiased estimates, however, the regressors in the model must be uncorrelated with that random variable; otherwise the estimated effects of those regressors will be inconsistent.

Unlike a fixed effects approach, random effects estimation does not discard variation across individual units. The additional information inherent in the between-unit variation implies several advantages for the random effects approach over the fixed effects approach. One advantage is smaller sampling variability and thus narrower confidence intervals. In addition, the random effects method can estimate the effect of measured causes that do not vary over time – something the standard fixed

effects method cannot do, as we noted earlier. Moreover, with respect to measured causes that do vary over time, the random effects method allows coefficients to vary across individuals. This is done by inserting β_i in place of β in Eq. (7.6) and assuming that β_i is a vector of normally distributed random variables with a common mean and variance (Allison 2005: 27).

The virtues of the random effects method come at a potentially steep price, however. The random effects method assumes that the time-invariant individual differences are drawn from a random variable μ , *rather than treating them as fixed values*. Therefore, it does not automatically remove the effects of the time-invariant causes that are bundled up in the μ_i term – and thus does not enjoy the aforementioned fixed effects advantage. Absent that feature, the random effects method must – like most methods common to the social sciences – *assume* that the unmeasured causes in μ_i are uncorrelated with measured causes. That is often a difficult assumption to make and, if it is not true, the results will be subject to omitted-variable bias.

Ideally we would like a method that retains the fixed effects advantage – sweeping away the effects of all unmeasured time-invariant confounders – while also adding the attractive features of the random effects approach. To some extent this is possible by using random effects models that are centered in the way described in the next section.

Hybrid Model: The Centered Random Effects Model

We make three points in this section. First, fixed effects models can be estimated by centering each unit around its mean. Second, random effects models can be estimated by centering in the same manner. Third, centered random effects models possess the fixed effects advantage of eliminating the effects of unmeasured time-invariant causes. As a result, centered random effects models replicate the results of fixed effects models for regression coefficients and their standard errors, while being more flexible than the standard fixed effects model.

Centering to Estimate Fixed Effects Models

To move to a consideration of centering around the mean, we note first that fixed effects models can be estimated using change scores instead of dummy variables. This is easiest to see in the case of two observations for each person. Substituting $t = 1, 2$ into Eq. (7.5) yields these equations for time 1 and time 2, respectively:

$$Y_{i1} = \alpha_1 + \mu_i + \beta X_{i1} + \varepsilon_{i1} \quad (7.7a)$$

$$Y_{i2} = \alpha_2 + \mu_i + \beta X_{i2} + \varepsilon_{i2} \quad (7.7b)$$

Subtracting (7.7a) from (7.7b), we have:

$$Y_{i2} - Y_{i1} = (\alpha_2 - \alpha_1) + \beta(X_{i2} - X_{i1}) + (\varepsilon_{i2} - \varepsilon_{i1}) \quad (7.8)$$

Observe that Eq. (7.8) can be estimated by regressing change in Y on change in the X s. The y -intercept is the difference in the y -intercepts at time 1 and time 2, and the β s are the effects of the change in X on the change in Y . The μ_i have differenced out, meaning that the model alleviates omitted-variable

bias arising from time-invariant confounders. Moreover, if ε_{i1} and ε_{i2} both satisfy the assumptions of the standard regression model, their difference ($\varepsilon_{i2} - \varepsilon_{i1}$) will satisfy those assumptions as well, and Eq. (7.8) can be estimated using ordinary least squares.

In estimating the model in (7.8), suppose that we replaced the values of the Y s and the X s with values that were centered around their respective unit-specific means. More formally, let $x_{it} = X_{it} - \bar{X}_i$ be called the deviation score. Then, Eq. (7.8) could be rewritten as:

$$y_{i2} - y_{i1} = (\alpha_2 - \alpha_1) + \beta(x_{i2} - x_{i1}) + (\varepsilon_{i2} - \varepsilon_{i1}) \quad (7.9)$$

In the case of two observations per unit, then, it is easy to see that, instead of estimating the fixed effects model by regressing change in Y on change in the X s, we would get the same results by regressing differences in unit-specific deviation scores for Y on differences in unit-specific deviation scores for the X s. This is the case because the difference in the deviation scores is the same as the difference in the values themselves, for example, $x_{i2} - x_{i1} = (X_{i2} - \bar{X}_i) - (X_{i1} - \bar{X}_i) = X_{i2} - X_{i1}$.

Although there is nothing to be gained by using the deviation score fixed effects method of Eq. (7.9) in place of the simple change score fixed effects method of Eq. (7.8), the deviation score method becomes preferable when there are multiple observations for each unit. Moreover, readily available software packages make it easy to estimate fixed effects models using deviation scores. In SAS, for example, this can be accomplished by using the ABSORB command in PROC GLM (Allison 2005); in STATA, if the data are in the wide form, unit-specific means across multiple observations of a given measure can be obtained by using the “egen” command combined with the “rowmean” option.

In general, then, with repeated data points, fixed effects models are estimated using unit-specific deviation scores. That point bears on our discussion here because there is nothing to prevent one from also using unit-specific deviation scores in the estimation of random effects models – as we elaborate in the next section on the centering of random effects models.

Before that discussion, however, there is a final point to be made about fixed effects models: In differencing out the effects of constant unmeasured variables, the fixed effects model in (7.8) also differences out the effects of constant *measured* variables. As noted earlier, this is problematic when the causal variables of interest are time-invariant. It is possible nonetheless to use the fixed effects model to estimate whether or not the *effect of a constant trait has changed* over time. To see this through notation, consider that the term $\beta(X_{i2} - X_{i1})$ in Eq. (7.8) assumes that the effect of the X s did not change over time. To permit the effects to change, we replace $\beta(X_{i2} - X_{i1})$ with the term $\beta_2 X_{i2} - \beta_1 X_{i1}$ and rewrite Eq. (7.8) as:

$$Y_{i2} - Y_{i1} = (\alpha_2 - \alpha_1) + \beta_2(X_{i2} - X_{i1}) + (\beta_2 - \beta_1)X_{i1} + (\varepsilon_{i2} - \varepsilon_{i1}) \quad (7.10)$$

To illustrate, suppose we want to investigate the effect of gender on earnings in the United States. Consistent with Eq. (7.10), we regress the change in earnings on the determinants of earnings measured at time 1 (X_{i1}) and on the change in those determinants from time 1 to time 2 ($X_{i2} - X_{i1}$). Although we cannot estimate the causal effect of gender because the term $X_{i2} - X_{i1}$ drops out for gender, we can determine whether the gender effect has *changed* over time, since gender at time 1 (X_{i1}) does not difference out in Eq. (7.10) – in other words, even though we cannot estimate either β_1 or β_2 for gender, we can estimate whether they are different. So if we know that males have an earnings advantage, and we code the gender dummy variable as 1 for men and 0 for women, a positive coefficient for the gender dummy would indicate a growing gender effect and a negative coefficient would indicate a declining gender effect. In that way, we can at least determine whether the gender gap is widening or narrowing, which may have important substantive implications.

Centering to Estimate Random Effects Models

To retain the ability to remove unmeasured time-invariant confounders while allowing for variation across individuals, suppose we include both the within- and between-person variation in the time-varying predictors as *separate predictors* in our regression model. In other words, we decompose the time-varying predictors into two parts – a deviation score for the within-person variation, and a mean value for each person to capture between-person variation – and enter those components as separate predictors in the regression equation. The regression coefficients for the within-person component will be identical to the coefficients in the standard fixed effects model, thereby retaining the fixed effects advantage. At the same time, because the model incorporates between-person variation, this “hybrid” model retains the virtues of the random effects approach.

Note that we retain the fixed effects advantage by differencing out the unit-specific means from each time-varying predictor. In the literature on multilevel models, the subtraction of unit-specific means is called *group-mean centering* (Raudenbush and Bryk 2002). Because the term “unit” can refer to individuals or to organizations such as schools or to spatially defined areas such as census tracts or cities, unit-specific means could exist at different levels of aggregation, with some units possibly nested within others. Regardless of the specific level of analysis, because each predictor is expressed as the deviation from its unit-specific mean, by centering around the unit means the model differences out the effects of all unmeasured time-invariant causes at that level, thereby eliminating that source of bias at that level.

In estimating random effects models with group-mean centering, the first step is to create a data set consisting of unit-specific means, and deviations from those means, for each of the time-varying predictors in the model (there are software programs that will do this automatically). This procedure is applied only to the predictors, not to the dependent variable. As noted above, the deviation scores capture the within-person variance and the means provide the between-person variance. Then we regress the dependent variable on the random variable μ and on the means and deviation scores for the X s (see Raudenbush 2009, equation 7).

This type of hybrid fixed effects-random effects model gives the same estimates for the time-varying predictors as the fixed effects model does (Allison 2005; Raudenbush 2009). Why, then, use this method in place of the standard fixed effects method? First, the hybrid model allows for the estimation of the effects of time-invariant as well as time-varying predictors. Second, by permitting random intercepts for each person, the hybrid model gives estimates of between-person effects. This is important because the between-person coefficients can be compared with the corresponding within-person coefficients to test whether the person-specific effects (the μ_i) are in fact independent of the time-varying causes in the model – the key assumption of the random effects model. If there are no differences (so the assumptions of the random effects model hold), then the more flexible random effects method is preferred over the fixed effects method. By using the hybrid model, then, we can determine whether it is appropriate to use the more efficient random effects method in place of the fixed effects method.

Types of Observable and Unobservable Causes

As we have seen, the fixed effects approach is most appropriate when the causal variables of interest change over time and the unmeasured causes are stable. In contemplating the use of a fixed effects model for a particular analysis, then, it is useful to begin by thinking about the nature of the likely

observable and unobservable causes (see also Chap. 3 by Freese and Kevern, this volume). Two basic conditions should be satisfied:

1. The causes we are most interested in should change over time.
2. By contrast, unobservable causes should be constant over time.

With regard to the first condition, we have already noted that standard fixed effects models cannot be used to estimate the effects of ascribed characteristics such as gender or race. This problem is not fatal, however, since the effect of such causes can be estimated using hybrid models. In reality, the problem researchers are more likely to face with fixed effects analysis is a limited amount of change – rather than the absence of change – in the key variables of interest. The issue is one of variance, as variables must, of course, vary to determine their effects. The fixed effects coefficient is based on within-unit variance, that is, on change over time. If we want to estimate the effect of schooling on income in the United States, for example, we must remember that the vast majority of adults in the United States complete their formal education by age 30, so fixed effects analyses using a sample of largely middle-aged adults is likely to understate the effect of schooling on (say) income because incomes generally continue to increase in the post-schooling years. Although the lack of variance on the predictor is clear in this example, it might be less obvious in other instances, and researchers should be alert to the potential problem.

We have addressed the second condition earlier as well. The point we want to stress now is that the assumption of constant unobservables is very often quite defensible. As Allison (2005) points out, the confounders that are hardest to measure are often time-invariant, or nearly so. In any case, in nonexperimental research, the assumption – embedded in random effects models – that the unmeasured causes are uncorrelated with measured causes generally requires a much bigger leap of faith than the assumption that those causes are not changing over time (as assumed in fixed effects models). Although the operative variance in fixed effect models is smaller, it is often more telling, since it has been purged of the enduring effects of the unmeasured – and often hard-to-measure – time-invariant traits of the units under consideration.

Illustrations

Determinants of Wages in the United States

We illustrate the fixed effects, random effects, and hybrid models using wage data from a portion of the National Longitudinal Survey of Youth (NLSY79). In conjunction with the Bureau of Labor Statistics (BLS), data collection for the NLSY79 began in 1979 on a nationally representative sample of 12,686 young men and women who were 14–22 years old. We focus here on a subset of 10 survey waves collected annually from 1983 to 1992, when the respondents were ages 18–35. Our dependent variable is the logged hourly wage rate for a respondent's current or most recent main job.²

To provide a baseline for the fixed effects, random effects, and hybrid models, we begin with results from a simple OLS regression with basic controls. Suppose we modeled the logged hourly wages for young adult workers in the United States as a function of job tenure (number of weeks on current job, logged), hours worked per week, age and age squared, and years of education, with dummy variables for race, urban residence, gender (male = 1), marital status (married = 1), parental status (living with child under age 10 = 1), and work status (full-time = 1). We also include nine dummy variables for time, one each for years 1983 through 1991.

²To adjust for inflation across the 10 years of data collection, all wages are standardized using the consumer price index to obtain wages in 1983 dollars.

Table 7.1 Predicting logged hourly wage rate across model specification: NLSY79 (1983–1992)

	OLS regression		Random effects	Fixed effects
	Cross sectional (1992)	Pooled W/time dummies		
	Model 1	Model 2	Model 3	Model 4
Job tenure (in weeks, logged)	0.97*** (.04)	0.82*** (.01)	0.56*** (.01)	0.44*** (.01)
Hours worked per week	−0.05*** (.01)	−0.06*** (.002)	−0.06*** (.002)	−0.06*** (.002)
Full time worker	2.81*** (.24)	2.86*** (.07)	2.02*** (.06)	1.68*** (.06)
Age	1.18 (.77)	1.07*** (.06)	1.26*** (.06)	1.28*** (.08)
(Age) ²	−0.02 (.01)	−0.02*** (.001)	−0.02*** (.001)	−0.02*** (.001)
Years of education	0.76*** (.03)	0.63*** (.01)	0.68*** (.01)	0.77*** (.03)
Marital status	0.71*** (.14)	0.67*** (.04)	0.51*** (.04)	0.41*** (.05)
Children under age 10	0.24 (.14)	−0.12** (.04)	−0.11* (.05)	−0.09 (.05)
Urban residence	1.62*** (.16)	1.43*** (.04)	0.96*** (.05)	0.60*** (.06)
Male	2.16*** (.13)	2.18*** (.04)	2.43*** (.07)	— —
Race (white omitted)				
African American	−1.81*** (.15)	−1.29*** (.04)	−1.30*** (.08)	— —
Hispanic	−0.48** (.17)	−0.28*** (.05)	−0.15 (.09)	— —
Intercept	−1.75 (1.19)	−1.31*** (.08)	−1.46*** (.08)	−1.34*** (.17)
R ²	0.27	0.26	—	—
Variance components				
Intercept	— —	— —	0.10*** (.00)	— —
Residual	— —	— —	0.15*** (.00)	— —

Notes: * $p < .05$; ** $p < .01$; *** $p < .001$; All models (other than Model 1) include dummy variables for survey wave (wave 10 omitted); Sample size for Model 1: 7,155 respondents; Sample size for Models 2–4: 83,725 person observations clustered in 11,743 respondents (average of 7.1 waves per respondent); All coefficients and standard errors multiplied by 10

With the NLSY data, we could have added more predictors, but a relatively basic model serves our purposes here. Table 7.1 presents estimates for the predictors described above for two OLS regressions. First, to show what the results would look like if we regressed wages on the predictors for a given year, we regressed hourly wages in 1992 on job tenure, hours worked per week, and the other predictors in 1992 (model 1 in Table 7.1). Cross-sectionally, variance in the predictors across individuals accounts for 27% of the variance in log hourly wages across individuals. Because of the relatively large sample, almost all predictors have statistically significant effects; the only exceptions are age and age squared and resident children under age 10. Other things equal, full-time workers earn more than part-time workers, and workers who have been in their current job longer tend to earn

higher hourly wages than those who were recently hired. On the other hand, controlling for full-time versus part-time status and the other variables in the model, those who work longer hours tend to have diminished hourly wages.

Education has a significant positive effect on wages. Workers who are married tend to earn more than those who are not, with the effect of being married being about the same as the effect of an additional year of education. Men tend to earn more than women and whites tend to earn more than Hispanics and (especially) African Americans. Urban residence has a positive effect on earnings; in fact, the OLS coefficients in this cross-sectional model indicate that the wage gap between urban and rural workers is almost as large as the wage gap between whites and African Americans. In this OLS model, women need nearly three more years of education than comparable men to match their wages, and African Americans need over two more years of education than comparable whites to match their wages.

The second model in Table 7.1 also employs OLS regression, but this time we use the pooled (i.e., panel) data, which allows us to make use of variance over time as well as variance across individuals. In this model, we include dummy variables for time (not shown) but not for individuals. Including dummy variables for individuals would create a fixed effects model, which we do not want to estimate yet, because our purpose is to create baseline estimates against which to compare the results of fixed effects and random effects models. Given the larger sample size we expect the coefficients in the second model to attain statistical significance, and they all do.³

The switch from cross-sectional to panel data does not change the results dramatically. The coefficients for the three work variables – job tenure, hours worked per week, and full-time versus part-time work status – remain about the same, except that the effect of job tenure is somewhat attenuated. The effects of marital status and gender are about the same. The effect of urban residence declines by about 12%, and the effect of education declines by about 17%. The coefficient for resident children under age 10 is now negative (as opposed to zero) and statistically significant, but the effect is marginal. The most noteworthy changes involve the wage gaps between whites and minorities, which in the panel data are about 30% narrower for African Americans and whites and about 40% narrower for Hispanics and whites.

Even if we added many more predictors to these OLS models, we could never realistically claim that we have included every difference between workers that could affect their earnings. What we want to know, then, is how well the OLS findings from the panel data hold up when we add the person-specific term μ_i either as a random variable (as in the random effects model) or as a fixed value (as in the fixed effects model). Those results are also reported in Table 7.1, as models 3 and 4 respectively.

Three patterns immediately stand out. First, adding the μ_i term affects the results substantially. The results for models 3 and 4 differ from the results for model 2 more than the results for model 2 differ from those in model 1. In other words, the OLS estimates for the panel data are closer to the OLS estimates for the cross-sectional data than they are to the fixed effects and random effects estimates for the panel data. This observation is noteworthy because it indicates that estimation method (fixed or random effects versus OLS) is more important here than the size and structure of the data (panel data over 10 years versus cross-section data for a single year).

Second, the estimated effects of the predictor variables are not always smaller in the random effects and fixed effects models than they are in the analogous OLS model. While some of the estimated effects are attenuated by the addition of the μ_i term, the estimates for other predictors are larger in the random effects and fixed effects models. Specifically, the random effects and fixed effects estimates are smaller for job tenure, full-time work status, marital status, and urban residence and larger for age and years of education.

³Because our purpose is to compare the results for the fixed effects, random effects, and hybrid models to a simple OLS baseline model, we do not adjust the OLS standard errors for clustering. In any case, adjusting the standard errors scarcely affects the results.

Table 7.2 Comparing random- and fixed-effects coefficients and models

	df	Chi-squared	<i>p</i> -value
Hausman test	17	1,207.22	<0.001
<i>Using centered scores</i>			
Job tenure	1	381.52	<0.001
Hours per week	1	2.07	0.150
Full time worker	1	195.43	<0.001
Age	1	6.31	0.012
(Age) ²	1	4.04	0.045
Years of education	1	20.19	<0.001
Marital status	1	9.59	0.002
Children under age 10	1	1.01	0.310
Urban residence	1	123.84	<0.001
Full model	9	1,236.27	<0.001

Note: Wald χ^2 tests of equality of coefficients. *p*-values of <0.05 suggest that coefficients are not equal and random-effects estimates are not consistent

Third, differences between the fixed effects estimates and the pooled data OLS estimates are consistently larger than the differences between the random effects estimates and the pooled data OLS estimates. Thus, when coefficients shrink with the addition of the μ_i term, they *decrease more* for the fixed effects; and when they increase with the addition of the μ_i term, they *increase more* for the fixed effects model. For example, the estimated effect of education on wages is 0.63 in the OLS panel model, 0.68 in the random effects model, and 0.77 in the fixed effects model, whereas the effect of job tenure on wages is 0.82 in the OLS model, 0.56 in the random effects model, and 0.44 in the fixed effects model.

The fundamental question becomes, which estimates most accurately represent *causal effects*? Consider the impact of job tenure on wages. If we suppose that workers who move from job to job tend to have unmeasured traits that differ from the traits of workers who stay where they are – perhaps those who jump from job to job have a weaker work ethic, or do not get along well with others – then the cross-worker comparison of the wages of short-term and long-term workers could be a misleading gauge of causal effects because it fails to separate out the difference-in-workers effect from the length-of-time-in-a-job effect. To find the true length-of-time-in-a-job effect, it would seem more appropriate to rely on variance within workers, as the fixed effects model does. By ignoring between-worker variance, we of course reduce the power of our models, but statistical power is not a major concern here, given the size of our sample.

More generally, how do we decide between the models? While it is clear that, in this case, the panel models are superior to the cross-sectional OLS model, and the fixed and random effects models are superior to the OLS panel model, there is still the choice to be made between the fixed and random effects methods. Should we assume, as in the random effects approach, that u is independent of the covariates? To answer this question empirically, we employ two approaches for comparing fixed and random effects models, one based on the Hausman test and the second utilizing the hybrid model outlined above. Both of these approaches, summarized in Table 7.2, are estimated as Wald χ^2 tests with $k - 1$ degrees of freedom.

Perhaps the most widely used, the Hausman specification test (Hausman 1978) compares an estimator known to be consistent when unobserved person-specific differences are time-invariant (the fixed effects estimator) with an estimator that is efficient under the null hypothesis that those unobserved differences across persons are orthogonal to the predictor variables (the random effects estimator). The Hausman test compares results from the fixed and random effects models to determine if there is sufficient evidence to reject the null hypothesis that the unobserved person-specific

differences are orthogonal to the regressors in the model. If we fail to reject the null hypothesis, then we assume that the unobserved heterogeneity is uncorrelated with the regressors, and thus that the random effects estimates are consistent. Otherwise, if we reject the null hypothesis – as we do in Table 7.2 – then we conclude that the random effects estimates are not consistent, and we reject them in favor of the fixed effects estimates.

For an alternative to the Hausman test, we can estimate the previously described hybrid model that contains both fixed and random effects estimators. The advantage of this approach is that it enables a direct comparison of estimates, coefficient by coefficient. When the wage model is estimated in this fashion, we find that the fixed and random effects coefficients in fact are significantly different for most of the predictors; the only exceptions are hours worked per week and resident children under age 10 (Table 7.2). Also, as in the Hausman test, we can test all of the time-varying variables simultaneously (final line in Table 7.2). When we do this, we find that the two tests point to the same conclusion (note the similar Chi-squared values). In the example under consideration, then, there is sufficient evidence that the random effects model should be rejected in favor of the fixed effects model.

Family Size and Children's Intellectual Development

For a second example, it is instructive to examine the results of an exchange in the April, 1999, issue of the *American Sociological Review*. In the lead article in that issue, Guo and VanWey (1999) apply fixed effects methods to investigate whether a child's intellectual development is affected by the number of siblings in a family. As they note, there is an extensive and contentious literature on that issue, stemming from the observation that children from larger families tend to have lower test scores and acquire less schooling. What is unclear, however, is whether family size has a *causal* effect on intellectual development. Perhaps it is the case that parents who choose to have more children are inherently different in a way that also affects motivation for schooling, so their children would have had lower test scores (the dependent variable in the Guo-VanWey study) regardless of number of siblings. This is precisely the sort of question that fixed effects models are well positioned to address, since the fixed effects models can sweep away the confounding effects of hard-to-measure persistent differences between couples who choose large families versus those who choose to have fewer children.

Contrary to earlier findings (e.g., Blake 1981; Downey 1995), the family fixed effects models of Guo and VanWey show limited support for the “dumber by the dozen” claim (Zajonc 1975). If we accept the Guo-VanWey results, the obvious implication is that previous findings supporting the claim that more children lead to lower test scores were distorted by omitted-variable bias. In their commentaries, Phillips (1999) and Downey et al. (1999) note several limitations of the Guo-VanWey study, and Downey et al. (1999) offer alternative explanations for the results presented by Guo and VanWey. We note that these criticisms are quite consistent with some of the issues discussed earlier in this chapter. One criticism, for instance, is that the results are based on small incremental changes in family size over time. This is the problem of limited within-unit variance which, as noted earlier, is not unusual in fixed effects analyses. A second potential problem raised by Downey et al. 1999 (and also addressed earlier in this chapter) is that fixed effects models remove only the effects of *time-invariant* unmeasured confounders. The failure to sweep away the effects of unmeasured *changing* family characteristics could be problematic if these characteristics change as additional children are added to the family. Third, the fixed effects models utilized by Guo and VanWey do not control for differences between siblings that might affect decisions parents make about whether to have more children (“difficult” children, for example, might discourage further childbearing). If these temperament differences are associated with differences in children's academic achievement, this

would bias the fixed effects estimates of family size on educational achievement. Finally, Downey et al. (1999) argue that the results of Guo and VanWey miss most of the family size effect because their sample omits closely spaced siblings, where we would expect the strongest size effects (a charge that Guo and VanWey dispute in their rejoinder).

In a subsequent study based on Norwegian data, Black et al. (2005) report findings that, consistent with those of Guo and VanWey (1999), undermine the notion that family size has a causal effect on children's academic development. Although Black et al. (2005) investigate the effect of family size on completed schooling rather than on test scores, their study is nonetheless noteworthy because of its magnitude. Using administrative files consisting of all children and parents aged 16–74 at some point during the interval 1986 to 2000, the authors matched individuals to their children, resulting in a population of 1,427,100 children from 647,035 families. To remove children who had not completed their schooling, the analysis is restricted to those who were at least 25 years of age in 2000. With a sample this size the authors are able to overcome many of the limitations of prior analyses of family size effects. There is, for example, enough statistical power to separate the effects of birth order from the effects of family size, a problem that had vexed prior studies. More to the point here, because a twin birth occurs in about one of every 65 families in this population, the authors were able to investigate the causal effect of family size by using the birth of twins as a source of exogenous variation in family size. The coefficient is negative and statistically significant, but negligible in magnitude. In line with the findings of Guo and VanWey (1999) for the United States, then, Black et al. (2005) find that family size has little causal effect on academic achievement in Norway.

Estimating the Effect of a Particular Cause: Incarceration and Wages

As a final illustration, we estimate the causal effect of incarceration on wages by adding a dummy variable for incarceration to our previous fixed effects model of wages. This application of the fixed effect model differs from the previous examples because we are interested in only one effect – the effect of incarceration – *and* the effect of incarceration pertains only to a small subset of the population. This illustration, then, touches upon two important practical issues: the use of fixed effects models to estimate the “effect of a cause” (Morgan and Winship 2007) and the issue of severely limited variance on the focal variable.

Because more than 700,000 individuals are released from American prisons each year (Glaze 2011), the reintegration of ex-inmates into American society has become a pressing national issue. We know that ex-inmates are more likely to be unemployed and to earn lower wages when they are employed (Pager 2003; Western 2006). However, the question of how much these differences reflect the *causal effect* of being in prison is not an easy one to answer because inmates are not a random sample of American adults. Compared to the rest of the U.S. adult population, a prisoner is more likely to be male, young, poor, unemployed, a member of a racial or ethnic minority, and to have a low level of education (Western 2006: 16–18).

To provide a fixed effects estimate of the causal effect of incarceration on wages, we added a dummy variable coded 1 for all years *after* an individual is released from prison. Correctional confinement is measured in the NLSY using an annual residence item, and our measure of ex-inmate status is coded 1 for all survey waves after a respondent is last interviewed in prison. It is coded 0 for all pre-prison waves and for all waves among the never-incarcerated respondents. For our purposes here, we drop respondents from the regression models during the waves they were interviewed while incarcerated. Given that most respondents in the NLSY have never been incarcerated, the incarceration values consist of a string of zeros for most respondents. In this case, then, the use of a fixed effects approach represents a classic example of limited variance on the variable of interest.

Table 7.3 Fixed effects estimates of the effect of incarceration on wages: NLSY79 (1983–1992)

	Fixed effects model	
	All respondents	Only ex-inmate sample
	Sample 1	Sample 2
Ex-inmate status	−0.65* (.26)	−0.58* (.27)
Job tenure (in weeks, logged)	0.45*** (.01)	0.38*** (.07)
Hours worked per week	−0.06*** (.00)	−0.03* (.01)
Full time worker	1.66*** (.06)	0.12 (.32)
Age	1.31*** (.08)	1.37** (.45)
(Age) ²	−0.02*** (.00)	−0.02** (.01)
Years of education	0.79*** (.03)	0.08 (.22)
Marital status	0.39*** (.05)	−0.08 (.32)
Children under age 10	−0.12* (.05)	0.25 (.28)
Urban residence	0.57*** (.06)	0.55 (.38)
Intercept	−1.39*** (.17)	−0.96 (.97)
Person-observations	83,240	2,601
Persons	11,730	499

Notes: * $p < .05$; ** $p < .01$; *** $p < .001$; Both models include dummy variables for survey wave (wave 10 omitted). All coefficients and standard errors multiplied by 10

Table 7.3 reports the results when the ex-inmate dummy is added to the fixed effects model in Table 7.1. Sample 1 in Table 7.3 utilizes the full sample of NLSY respondents. Consistent with Western (2002), we find that incarceration depresses wages, though perhaps not as much as one might expect. The estimated causal effect of incarceration is somewhat larger in absolute size than the effect of urban residence, and somewhat smaller than the effect of an additional year of education.

Yet, as discussed at various points in this chapter, the fixed effects approach focuses on individual change. This raises the issue of how best to estimate the effect of a single cause when, as here, most subjects contribute no variation on that cause. In the current analysis, how do our estimates change when we examine only respondents with a history of incarceration?

The second sample in Table 7.3 reports the fixed effects estimates when we restrict the analysis to ex-inmates. Although the new estimates draw upon dramatically fewer cases – 499 respondents (2,601 person-observations) as opposed 11,730 respondents (83,240 person-observations) – the coefficient for the incarceration variable remains statistically significant, and is similar in magnitude (Table 7.3). This reflects that fact that the original estimate of the incarceration effect (Table 7.3, sample 1) is based on the 499 ex-inmates, since those respondents are responsible for *all* the variation on incarceration. In other words, as specified in our model, the incarceration parameter represents within-person change in earnings attributable to incarceration. That effect can only be estimated among those with variance on incarceration status. Because those individuals are the same in models 1 and 2, there are only minor differences in the incarceration slopes across the two samples.

Moreover, because the ex-inmates supply all the variation for the incarceration variable, the standard errors for the incarceration coefficient are virtually the same for the two samples. Thus our conclusion regarding the causal effect of incarceration on wages is the same whether we include or exclude the 11,231 respondents who were never incarcerated. Our conclusions about the effects of the *other* predictors differ, not surprisingly, across the two samples. Based on the ex-inmates sample we would conclude that wages are determined only by job tenure, hours worked per week, and age, in addition to the incarceration effect; the other predictors in the model – education, full-time worker status, and so on – have no effect on wages in the fixed effects estimates for ex-inmates.

Extensions and Further Reading

In this chapter, we have introduced readers to the fundamentals of fixed effects, random effects, and hybrid fixed effects-random effects models by focusing on the general linear regression model for continuous dependent variables. We conclude our chapter on fixed effects models by noting extensions of the fixed effects and hybrid models to other types of dependent variables and pointing interested readers to some of the key literature describing fixed effects, random effects, and hybrid models in greater detail. Before doing so, however, we first point out a fixed effect method that most researchers are familiar with, and perhaps have even used, without realizing that it belongs to the fixed effects family of models.

Growth Rate Models

Rate of growth over a time interval often is calculated based on values at the end points of the interval. When that method is used, the rate of growth of Y from time 1 to time 2 is $\log Y_{i2} - \log Y_{i1}$ (where $\log$ denotes natural logarithm), so regressing growth rates on growth rates is in effect the regression of change scores on change scores.

Because this type of growth rate model regresses change on change, it is the same model as in Eq. (7.8) above, except that the variables in the growth rate model are logged. In both cases, the μ_i have been differenced out, sweeping away potential omitted-variable bias arising from stable confounders. Hence, like other members of the fixed effects family, growth rate models of this type enjoy the fixed effects advantage.

In addition to rate of change, growth rate models can also include initial level as a regressor. Recall that the parameter attached to X at time 1 is $\beta_2 - \beta_1$ in the fixed effects model (Eq. 7.10 above). By adding X at time 1 to the growth rate equation, then, we can determine whether the effect of the growth rate of a predictor on the growth of a dependent variable depends on the initial level of the predictor. This specification of the model can be especially useful for investigating the presence of ceiling or floor effects, or of scale economies or diseconomies.

Fixed Effects and Hybrid Models for Categorical Dependent Variables and Beyond

Fixed effects can be utilized with logistic regression models for categorical dependent variables, in Cox regression models for event history data, and in Poisson models for count data, as we now see.

Categorical Dependent Variables

The fixed effects estimator is sometimes referred to as the “within estimator” because it relies solely on within-unit change over time. Should we then include units that do not change over time? The question arose earlier with respect to a categorical predictor (incarceration), where incarceration can be viewed as a “treatment” that applies only to a subset of the population. The question arises in the case of categorical *dependent* variables because estimates will differ depending on whether we apply logistic regression to all the units, resulting in *population-averaged estimates*, or apply logistic regression only to the units that change, yielding *subject-specific estimates*.⁴ Indeed, unlike fixed effects estimation for continuous dependent variables – where there are multiple ways to obtain the same results – the options for subject-specific fixed effects estimates are limited when the dependent variable is categorical. One possibility is conditional logistic regression (Allison 2005, chapter 3), a method that yields subject-specific estimates by in effect limiting the sample to those individuals who changed on the response variable over the period of observation. In these fixed effects models, then, information is lost in two ways: first by restricting the analysis to within-unit change and second by excluding units that did not change on the response variable.

Event History Data and Analysis

Event history analyses investigate the occurrence and timing of events. The events can be either repeatable (e.g., divorce) or nonrepeatable (e.g., death). Event history data consist of the record of events for individuals over a fixed interval of time, so we know when events occurred (or approximately so) and their sequence. Because some individuals contribute multiple spells, treating each data point as a separate and unique observation, when they are in fact related, results in downwardly biased standard errors. To solve this problem, Allison (1996) suggests the application of Cox’s partial likelihood method with each individual treated as a separate stratum. The result is a fixed effects model that alleviates both the nonindependence-of-observations problem as well as the problem of omitted-variable bias due to unmeasured time-invariant predictors. As with other fixed effects methods, the partial likelihood fixed effects approach can result in a significant loss of statistical power because it is restricted to within-unit variance.

Count Data

Fixed effects methods are also available for panel data where the response variable is based on counts that are highly skewed to the right. For instance, criminologists studying arrest data often find that number of arrests in a given year is a highly concentrated event, with a few individuals accounting for most arrests (Wolfgang et al. 1972). Undergraduate class size in many research universities very often is also highly skewed. This results in the well-known “size dilemma” in that, with many small classes and a few huge ones, most students are experiencing huge classes, even though large classes are the exception. Poisson models and negative binomial models have been developed to handle such skewed response variables. Fixed effects and hybrid models with panel data are available for both Poisson and negative binomial distributions. As in the hybrid models described earlier, the hybrid models

⁴A *subject-specific* coefficient estimates the change in Y for a particular individual if the predictor variable were increased by one unit. A *population-averaged* coefficient estimates the change in Y for the whole population if the predictor variable were increased by one unit for everyone. The two estimates are equivalent for linear models, but not for nonlinear models, such as logistic regression models (see Allison 2005, chapter 3).

here are formed by including two terms – a unit-specific mean, and a deviation from that mean – for each of the predictors. Allison (2005, chapter 4) describes these models, along with SAS programs to estimate them.

Suggestions for Further Reading

In our view, one of the best places to start is Paul Allison's *Fixed Effects Regression Methods for Longitudinal Data Using SAS* (2005). This excellent book contains separate chapters on fixed effects and hybrid models for categorical dependent variables (chapter 3), for count data (chapter 4), and for event history analysis (chapter 5). Each chapter provides plenty of examples rooted in actual data, along with the SAS commands used to generate the output. Even those not proficient in SAS can gain much from this reader-friendly book, as it provides a nice balance of theory and application. Although the SAS programs in the book will become dated over time as the software changes, this book will remain an important reference for social researchers interested in fixed effects methods and applications.

Another user-friendly, but shorter, exegesis of fixed effects models is available in Halaby (2004). Halaby's review chapter is a good source for readers seeking more detailed information on tests for comparing fixed effects and random effects results, and on the application of fixed effects approaches in models with lagged dependent variables.

Firebaugh's *Seven Rules for Social Research* (2008) describes fixed effects models in the context of the fifth rule, "Compare like with like." Fixed effects models compare like with like by matching units to themselves at different points in time. In line with Allison (2005), Halaby (2004), and the current chapter, this treatment of the fixed effects approach stresses its potential for alleviating omitted-variable bias.

Hsiao (2003) provides a more detailed econometric treatment of fixed and random effects models in his *Analysis of Panel Data*. The book is logically organized, beginning with analysis of covariance and proceeding to simple fixed and random effects models before addressing the complications added by heteroscedastic and autocorrelated disturbances. The book also includes chapters on dynamic models with variable intercepts, simultaneous equation models (including instrumental variables approaches), variable-coefficient models, and models for discrete response variables, truncated and censored data, and incomplete panel data.

Raudenbush and Bryk (2002) discuss fixed effects and random effects models in the context of multilevel models in their book *Hierarchical Linear Models*. The book, which is widely used in graduate seminars on multilevel modeling, goes into considerably more detail than space permits in the present treatment. Readers may find the book an appropriate extension of our chapter. For example, the book devotes an entire chapter to hypothesis testing, which affords the readers an opportunity to learn about a variety of robustness and model specificity tests. The book also devotes one chapter to individual change and a different chapter to organizational change. This allows the authors to direct considerable attention to some of the nuanced differences between fixed, random, and hybrid effects models, depending on the population of interest.

Of recent substantive applications of fixed and random effects models in sociology, we note two that are especially useful because of the templates they provide for researchers. Teachman (2011) demonstrates how to implement random and fixed effects techniques to approximate hazard models, that is, models where the researcher is interested in the time that it takes for a particular event to occur. Teachman argues that if the data are treated as discrete-time (i.e., pooled data in which the event in question either does or does not occur at each interval), the resulting database can be modeled using any procedure appropriate for binary outcomes (e.g., logit). Teachman combines this reasoning with multilevel modeling techniques for nominal data to show how to construct a multilevel discrete-time

hazard-rate model to examine marital dissolution, a repeatable event. Although the example he uses involves events clustered within individuals, he notes that the same logic applies to other types of clustering, such as individuals clustered within geographic units. Teachman explains in detail why results in this framework can vary across different model specifications: logistic regression, random effects model, fixed effects model, and a Generalized Estimating Equations (GEE) model. He also includes both the SAS and STATA code used to estimate the models he presents.

Readers who prefer to work within a structural equations framework should refer to Bollen and Brand's (2010) discussion of a general panel model that includes fixed effects and random effects models as special cases (see also Chap. 15 by Bollen and Pearl, this volume). The key insight is that random and fixed effects specifications can be modeled by imposing restrictions within a general structural equations model (SEM) framework. Using this strategy enables researchers to specify and test a sequence of nested models that can be compared using likelihood ratio and fit statistics that are readily available in standard SEM software. The authors present the classic random and fixed effects models as path diagrams and outline a series of steps researchers can use to build up to those models. Readers who are unclear about the nature of the relationship between time-varying and time-stable observed and unobserved variables will find these figures very useful. Bollen and Brand also discuss how their general SEM panel model can incorporate lagged effects, and they detail a number of tests of model fit that can help researchers decide which model specification most closely represents the data under consideration.

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